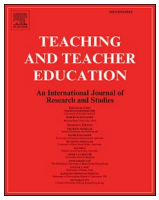
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Review article



A systematic review of teacher engagement in digital multimodal composing: Bridging the "last-mile" towards learner outcomes

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ABSTRACT

Digital multimodal composing (DMC) has become a central literacy practice in language education, yet research on teacher engagement remains conceptually fragmented and weakly linked to learner outcomes. This systematic review synthesises 51 empirical studies (2015–2025) that operationalise teacher engagement in DMC and examine its relationship with learner outcomes. Teacher engagement is organised into five interrelated dimensions: Orientation, Knowledge, Design, Assessment, and Enactment (OK-DAE), functioning as an analytic heuristic for mapping the literature. Using descriptive and co-occurrence analyses, alongside R (v. 4.5.1) for data visualisation and quantitative mapping, results show that Design and Enactment dominate the evidence base, while Orientation and Assessment are under-examined, and Knowledge is rarely examined. Moreover, only two studies report statistically tested learner outcomes alongside explicit teacher-engagement evidence. The findings reveal a persistent "last-mile" gap between pedagogical engagement and measurable learning progress, highlighting the need to strengthen how teacher engagement is conceptualised, measured, and supported.

1. Introduction

Digital multimodal composing (DMC) has become a core literacy practice in language education, grounded in multimodality and multiliteracies theories that foreground the orchestration of linguistic and non-linguistic resources for meaning-making (Hafner, 2015; Jewitt, 2008; Jiang, 2017; Kress, 2003, 2009; Tour et al., 2024). Implementing DMC in classrooms is not a single technique but a complex instructional activity shaped by teachers' knowledge, beliefs, task design and scaffolding, assessment of multimodal products, and day-to-day enactment under local constraints (Zhang & Yu, 2025b).

Drawing on existing theoretical constructs, we view teacher engagement with DMC not as a single act but as a multidimensional process involving five key interrelated dimensions identified in the literature: (1) orientations and beliefs toward DMC (e.g., teacher self-efficacy), (2) knowledge bases (e.g., pedagogical content knowledge [PCK], technological pedagogical content knowledge [TPACK]), (3) design, scaffolding, and orchestration practices in planning and running DMC activities, (4) assessment practices (criteria and rubrics for multimodal outputs and feedback routines), and (5) enactment and implementation fidelity under contextual constraints. These dimensions align with prior work on teacher engagement, stance continua, and knowledge for DMC, while recognising that engagement is developmental and context-sensitive rather than binary (Jiang et al., 2021; Leander, 2009; Lim & Polio, 2020; Louis & Smith, 1992; Mishra & Koehler, 2006; Schmidt, 2019; Shulman, 1986, 1987).

While belief-oriented constructs can motivate adoption, cross-system evidence cautions that generalised self-efficacy alone does not reliably translate into improved achievement without corresponding changes in enacted practice. This underscores the importance of causally active instructional mechanisms (modelling, rehearsal/feedback, prompts/cues, and action planning) and supportive conditions such as coaching and programme coherence that help convert intent into consistent classroom routines (Education Endowment Foundation, 2021; Jerrim et al., 2023; Sims & Fletcher-Wood, 2021). We therefore examine not only what teachers think but also what they know, design, assess, and do, and how these components are linked to learner outcomes in DMC settings (e.g., writing quality, engagement, performance).

Despite growing empirical interest, the existing evidence base remains largely descriptive, offering limited causal or analytical insight into how teacher engagement shapes learner outcomes. This lack of linkage constitutes a "last-mile" problem in DMC research, the persistent gap between pedagogical engagement and measurable learning progress (Durlak & DuPre, 2008; Lynch, 2024; McNaughton et al., 2018; OECD,

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2025). A further complication is that learner outcomes in DMC research are highly heterogeneous in both definition and measurement. Studies variably operationalise outcomes such as writing quality, engagement, or multimodal performance, often relying on context-specific or non-comparable instruments (e.g., Hafner, 2015; Jiang, 2017; Lim & Polio, 2020; Zhang & Yu, 2025a). This variability, combined with the frequent reliance on context-specific and often locally constructed assessment criteria (Hung et al., 2013; Lynch, 2024), makes direct correlational synthesis across studies methodologically challenging (e.g., Sims & Fletcher-Wood, 2021). Consequently, a systematic review is necessary at this stage to first map how outcomes are defined and measured before more robust correlational or causal designs can be meaningfully pursued. To address these gaps, the present review makes three contributions to the field. First, it identifies the "last-mile" problem in DMC pedagogy, foregrounding the scarcity of empirical linkages between teacher engagement and statistically tested learner outcomes. Second, to systematically map this fragmented landscape, we inductively synthesise multi-component principles from generic professional development research with specific DMC evidence to derive the OK-DAE framework. Third, by applying this framework, the review exposes structural imbalances in current research, offering a theoretically informed baseline to guide future empirical designs. Our analysis clarifies what has been measured, the extent to which teacher components relate to learner outcomes, and where validated instruments and multilevel models are still missing.

2. Literature review

2.1. Theoretical frameworks of DMC research

Digital Multimodal Composing (DMC) is rooted in the multiliteracies approach introduced by the New London Group (1996). This pedagogical framework emphasises developing students' abilities to effectively use various semiotic resources in a diverse and increasingly complex communication environment. At the core of DMC are the theories of multimodality (Kress, 2003, 2009) and multimodal literacies (Jewitt & Kress, 2003), which emphasise the integration of both linguistic and non-linguistic modes, such as language, images, sound, space, and gesture, into the process of meaning-making. From a social semiotic perspective, Kress (2009) argues that these modes, as distinct, socially shaped semiotic resources, each offer unique potentials for meaning-making and complement each other in constructing complex social meanings.

With the advent of the digital age, where text, image, sound, and video are integral to text construction (Jewitt, 2008; Kress, 2003), DMC has emerged as a cutting-edge literacy practice in language education (Hafner, 2015; Jiang, 2017). The essence of DMC lies in its recognition of the multimodal nature of contemporary texts, offering a comprehensive framework for language learning and teaching that addresses the complexities of digital communication (Hung et al., 2013). For instance, in an ESL classroom, students might create digital stories or video projects that combine text, images, and sound, thereby engaging multiple modes of communication and enhancing their learning experience.

Beyond foundational social semiotics, DMC research is increasingly informed by a synthesis of cognitive, sociocultural, and identity-oriented perspectives. Cognitively, DMC requires learners to actively orchestrate multiple semiotic resources. This process demands extensive planning for language formulation and complex rhetorical choices to represent knowledge across different modes (Lim & Polio, 2020; Zhang & Yu, 2025a). Socioculturally, DMC functions as a humanistic pedagogy that bridges conventional print literacies with students' out-of-school, everyday multimedia practices (Leander, 2009). By validating these practices, DMC empowers learners to use cultural resources from their lifeworlds to engage in authentic civic participation (Jiang, 2017; Tour et al., 2024). Consequently, engaging with DMC is not merely a technical skill but a transformative practice fostering learner agency and identity construction. As learners participate in digital "remix culture" to reach authentic audiences, they actively negotiate the expression of their unique "student voice" (Hafner, 2015). This reflects the sociological construct of student investment, where learners demonstrate deep investment in language practices that afford both identity validation and symbolic capital within increasingly fluid online and offline contexts (Darvin & Norton, 2015).

2.2. Teacher engagement in DMC: Addressing the "last mile"

Teacher engagement can be considered from multiple perspectives and is frequently invoked in the literature to explain variations in implementation quality and the success of pedagogical innovations (Durlak & DuPre, 2008; Louis & Smith, 1992; McNaughton et al., 2018). In its classic definition, teacher engagement refers to the psychological investment and effort teachers devote to enacting the knowledge, skills, and craft they aim to cultivate in learners (Louis & Smith, 1992). A large body of implementation research further shows that teachers' investment, participation, and enactment intensity strongly predict whether new instructional approaches are effectively implemented and whether pedagogical change can be sustained over time (Durlak & DuPre, 2008). In technology-mediated environments, instructional effectiveness similarly depends not on the digital tools themselves but on teachers' active design, scaffolding, and orchestration of learning processes, a phenomenon described by McNaughton et al. (2018) as conditional augmentation.

In the context of new literacy practices, Leander (2009) describes teachers' orientations as resistance, replacement, return, and remediation, capturing different stances towards integrating novel practices such as DMC. Extending this, Jiang et al. (2021) position teachers along an incidental-to-integral continuum, from peripheral use that merely supports traditional exercises to deep, curricular integration that values both linguistic and multimodal resources. Together, these frameworks extend the construct of teacher engagement beyond its original psychological definition to articulate its dynamic, developmental, and context-sensitive nature, shaped by teachers' beliefs, local policies, resource availability, and assessment regimes (Jiang et al., 2021; Leander, 2009).

2.2.1. Professional development and behavioural supports

The properties of teacher engagement mean that its development is sensitive to influence. Because of this, professional development (PD) is expected to play a critical role in enabling teachers to move along the engagement continuum. However, research shows that many teachers receive insufficient preparation in both the technical (e.g., editing, captioning accessibility) and pedagogical (e.g., genre, audience, multimodal orchestration) demands of DMC (Lim & Kessler, 2021; Lim & Polio, 2020; Zhang & Yu, 2025b). This reflects the insights from a recent critical reappraisal of PD evidence, which argues that the widely quoted "features" (sustained, collaborative, subject-specific, etc.) are often methodologically under-justified and conflate correlates with causes (Sims & Fletcher-Wood, 2021).

Instead, evidence points to the importance of causally active mechanisms (e.g., modelling, rehearsal with feedback, prompts/cues, goal-setting, action planning), which are more closely tied to changes in classroom practice (Education Endowment Foundation, 2021; Sims et al., 2025). In meta-analytic tests of more than 100 RCTs (randomised controlled trials), PD packages that combine more of these mechanisms and balance four purposes (to instil insight, motivate goals, teach techniques, and embed practice) produce larger gains in learner outcomes, albeit typically small-to-moderate in magnitude (Sims et al., 2025).

Nevertheless, even well-designed PD can stall at the point of classroom use due to what has been described as the "last-mile" gap—the difficulty of translating knowledge into consistent action, because

teachers often navigate limited attention, competing priorities, choice overload, and complex accountability pressures (Lynch, 2024). Behavioural economics has increasingly been used to address this issue by offering low-cost strategies to reduce implementation frictions in educational settings. These include checklists and decision trees, well-timed reminders, implementation-intention plans, simplified default options, and peer accountability structures (Education Endowment Foundation, 2021; Lynch, 2024). When embedded within coaching cycles or Professional Learning Communities (PLCs), these behavioural supports may help convert intent into consistent action, assisting teachers to move along the engagement continuum from incidental towards integral DMC.

The need for robust PD as well as implementation strategies is supported by McNaughton et al. (2018), who propose a Conditional Augmentation Hypothesis: the instructional power of digital tools is significantly amplified when mediated by deliberate teaching. Their findings suggest that mechanisms such as feedback, imitation, and agentive learning in digital environments yield more consistent and positive outcomes when teachers actively design and scaffold the learning experience. This reinforces the importance of teacher engagement not only as a continuum but also as a design-driven process, where behavioural supports and pedagogical choices interact to shape implementation fidelity.

Despite growing interest in PD designs that incorporate behavioural supports, there remains limited causal evidence on how specific mechanisms, such as rehearsal with feedback, reminders, and action planning, shift teachers from resistance or incidental engagement toward integral DMC, or how programme coherence and resource equity moderate this progression. Mechanism fidelity measures and A/B testing of "coaching only" versus "coaching-plus-nudges" models represent promising next steps.

2.2.2. Belief-oriented components and knowledge bases

While PD and behavioural supports shape how teachers enact DMC, effective implementation of DMC depends on teachers' pedagogical and technological knowledge, as well as their capacity to design, scaffold, and assess multimodal composing tasks (Jiang et al., 2021; Lim & Kessler, 2021; Lim & Polio, 2020; McNaughton et al., 2018). These capabilities draw on basic psychological and knowledge-based mechanisms, including teacher self-efficacy (TSE) (Tschannen-Moran & Hoy, 2001), pedagogical content knowledge (PCK; Shulman, 1986, 1987), and technological pedagogical content knowledge (TPACK; Mishra & Koehler, 2006). Empirical work further shows that teachers' TPACK, professional beliefs, and contextual support strongly shape how such knowledge is enacted in digitally mediated second language (L2) writing (Schmidt, 2019).

Among belief-oriented constructs, teacher self-efficacy has received particular attention; however, evidence on its predictive power is mixed. Large-scale evidence suggests that raising global TSE does not reliably translate into higher student achievement or durable practice change. A 23-system analysis using within-teacher, within-pupil contrasts in TIMSS 2015 reported near-zero associations between teachers' self-efficacy and achievement (and academic self-confidence), cautioning against PD that prioritises "feeling efficacious" over acquiring and automatising techniques (see Jerrim et al., 2023 for details). Instead, the literature urges a focus on adaptive expertise, defined as teachers' capacity to balance efficient routines with innovative adaptation under changing task and audience demands, cultivated through residency or clinical practice, joint design challenges, open mentoring, and frequent coached rehearsal (e.g., Moran et al., 2023).

2.2.3. System-level conditions

At the system level, teachers are more likely to progress from resistance to integral DMC when Continuous Professional Learning (CPL) architectures are coherent (school-embedded, content-focused, collaborative, and sustained) and when competency frameworks or micro-credentials make expectations visible. Equitable resourcing (devices, bandwidth, time, and coaching), especially in under-resourced schools, further reduces structural barriers. Such a system supports lowering the opportunity cost of adopting DMC and legitimises multimodal outputs through aligned rubrics and performance assessments (e.g., Zhang & Yu, 2025a). Most of this evidence is conceptual or policy-analytic rather than experimental and should be interpreted as hypothesised enabling conditions rather than established causal effects (OECD, 2025).

Synthesising current strands of research, teacher engagement with DMC appears most likely to deepen when PD (1) targets core mechanisms (e.g., modelling, rehearsal with feedback, prompts/cues, goal-setting, action planning), (2) embeds behavioural supports to reduce implementation friction, (3) builds adaptive expertise through in situ coached practice, and (4) is reinforced by system-level CPL and assessment alignment. However, empirical validation of this multi-level design as a pathway for moving teachers along the incidental-to-integral DMC continuum remains a critical gap addressed in the analytical focus of the present review.

2.2.4. Contextual variables

Beyond teacher orientations and expertise, contextual variables further shape how teacher engagement with DMC is enacted and sustained across settings. Task complexity matters for enactment: single-shot projects often confine design and feedback to a single outcome, whereas multi-task or sequenced multimodal assignments require ongoing orchestration, formative assessment, and coordination across modes, thereby making engagement more visible and sustained (Hafner, 2015; Lim & Polio, 2020).

Technological mediation also reconfigures teacher work; when digital tools are used, instructional effects are substantially amplified by deliberate teaching, through design, scaffolding, and feedback, rather than by technology alone (McNaughton et al., 2018). Recent developments in generative AI (GenAI) have further expanded the design and feedback possibilities within DMC tasks, introducing new instructional and psychological complexities (Ci & Jiang, 2025). Instructionally, as students prompt GenAI tools during composing, teachers must adapt tailored instructions and scaffolded guidance to navigate these new activity systems (Woo et al., 2025). Furthermore, evaluating GenAI outputs requires pedagogical attention to data quality, as models frequently generate fictitious references and rely on third-hand source materials (Spennemann, 2025). Psychologically, GenAI's capacity to mimic human cognition can threaten learners' perceived identity and uniqueness (Alessandro et al., 2025), while reconfiguring motivational constructs such as writing self-efficacy and self-regulated learning strategies (Liu & Zhang, 2025). Given these disruptive affordances and tensions, AI involvement acts as a critical contextual variable that systematically influences how teacher engagement manifests in multimodal instruction (Lee et al., 2025).

Educational level and language context (EFL/ESL/L1) mediate these dynamics via curriculum and assessment regimes that either afford or constrain sustained multimodal composing and feedback practices (Lynch, 2024; Sims & Fletcher-Wood, 2021). To account for these influences, the present review explicitly codes and reports four contextual descriptors (educational level, language context, task complexity, and AI involvement) so that subsequent analyses can distinguish descriptive prevalence from mechanism-relevant patterns and examine whether particular conditions are associated with stronger teacher–learner linkages in DMC. Mapping these features provides a coherent overview of where, with whom, and under what task and technological conditions teacher engagement in DMC has been studied so far, thereby clarifying the empirical landscape for subsequent analyses. Collectively, the mechanisms, system conditions, and contextual factors discussed above delineate the complex ecology within which teacher engagement in DMC operates. Taken together, the prior literature indicates what needs to be better understood in addressing the "last-mile" challenge in DMC research, namely, how teacher engagement mechanisms translate into

consistent instructional practice and, ultimately, improved learner outcomes.

2.3. The OK-DAE Framework of Teacher Engagement

To operationalise teacher engagement for this review, we employed an inductive process to derive a five-dimensional analytic heuristic, referred to as the *Orientation*, *Knowledge*, *Design*, *Assessment*, and *Enactment* (OK-DAE) framework (read as "OK day"). The conceptual origins of this framework lie in the intersection of the generic PD literature and the specific empirical evidence in the existing DMC research. Broad research on teacher professional learning establishes that effective pedagogical change requires a multi-component approach, demonstrating that isolated constructs are insufficient without corresponding knowledge and enacted practice (e.g., Jerrim et al., 2023; Sims & Fletcher-Wood, 2021). Existing empirical studies within the DMC literature have typically focused on documenting one or two dimensions in isolation, such as instructional design (Lim & Polio, 2020) or classroom enactment (Jiang et al., 2021), without a unifying structure to test their specific effects. Synthesising the components from the generic PD literature with the dimensions present across DMC studies yielded the five core dimensions of engagement. Rather than serving as a secondary layer of operationalisation, the OK-DAE framework directly operationalises this synthesis. It functions as a theoretically informed analytic lens to map how teacher engagement has been conceptualised and operationalised. It provides a structured representation of how teachers' beliefs and orientations shape their pedagogical knowledge; how knowledge informs instructional design and assessment practices; how design interacts with assessment; and how both culminate in enactment. Importantly, the primary function of the OK-DAE framework is to organise and interpret diverse conceptualisations of teacher engagement across studies.

The five dimensions operate as an interdependent network rather than discrete categories. We adopt the term Orientation to encapsulate teachers' values, efficacy beliefs, and motivational stances toward innovative pedagogy, capturing both cognitive and affective dispositions (e.g., Jiang et al., 2021; Leander, 2009). Knowledge encompasses professional, pedagogical, and technological knowledge bases that enable teachers to plan and facilitate learning (e.g., Mishra & Koehler, 2006; Shulman, 1986, 1987). Design represents the translation of beliefs and knowledge into structured learning tasks and multimodal experiences (e.g., Ci & Jiang, 2025; Hafner, 2015; Lim & Polio, 2020). Assessment provides formative feedback loops that both evaluate and refine design decisions, operating in reciprocal relation with them (e.g., Lynch, 2024; Sims & Fletcher-Wood, 2021). Finally, Enactment denotes the moment-to-moment implementation of tasks, scaffolding, and feedback in authentic learning contexts (e.g., Durlak & DuPre, 2008; McNaughton et al., 2018; Zhang & Yu, 2025b). Each dimension represents a necessary but not individually sufficient condition for sustained engagement. While these dimensions inevitably overlap in authentic practice (for instance, pedagogical knowledge is inherently embedded within instructional enactment), they function as methodologically separable lenses for the purpose of analytically mapping the literature. Together, they reflect interrelated dimensions that can be discussed in relation to pedagogical coherence and learner outcomes.

While the framework is analytically teacher-focused, it accommodates the possibility that more coherent enactment may relate to learner outcomes (e.g., engagement, performance), reflecting instructional effectiveness rather than a simple binary. In turn, learner outcomes may feed back into teachers' efficacy beliefs and motivational orientations and inform professional and pedagogical knowledge through reflective practice, together forming an iterative and mutually reinforcing cycle of sustained teacher engagement.

To emphasise this ecology, contextual supports, including professional learning, peer coaching, and institutional alignment, are positioned as outer-layer enablers that strengthen each dimension. These supports also indirectly influence teachers' orientations by fostering confidence and a sense of professional agency. The arrows in Fig. 1 denote influential rather than strictly causal relationships, illustrating iterative feedback and evolving interdependencies over time. While all dimensions are theorised to ultimately relate to learner outcomes, their conceptual pathways differ. Dimensions such as Design and Assessment are conceptualised as foundational antecedents that influence learner outcomes indirectly through the mediation of classroom Enactment. Consequently, Enactment is positioned as the primary interface shaping student performance. Conceptually, the OK-DAE framework bridges psychological investment and pedagogical action, offering a coherent lens for interpreting how teacher engagement develops and manifests across instructional design, assessment, and classroom practice.

2.4. Research questions

In summary, the comprehensive review of theoretical frameworks and empirical evidence highlights a critical need to examine how teacher engagement is defined, measured, and linked to learner outcomes in DMC contexts. While the OK-DAE offers a conceptual heuristic to organise these constructs, the extent to which current empirical research maps onto these dimensions remains to be synthesised. To address these gaps, this review investigates the following research questions.

1) What are the features of empirical studies (2015–2025) that have examined teacher engagement in DMC in relation to learner outcomes?
2) How is teacher engagement in DMC operationalised and measured, and with what evidence of reliability and validity?
3) To what extent are specific teacher engagement components linked to learner outcomes in DMC?

3. Methods and search strategy

3.1. Protocol and open science practices

The analytic protocol for this review was preregistered on the Open Science Framework (OSF; https://doi.org/10.17605/OSF.IO/32KQ4). To ensure transparency and replicability, the full search strategies, inclusion and exclusion decision rules, and the codebook are reported in this article. The complete screening logs, adjudication records, included study lists, and the coding datasets are accessible in the OSF project repository (https://osf.io/f4xg7).

3.2. Information sources and search strategy

Guided by a high-recall principle commonly adopted in systematic reviews, we designed a targeted search logic focusing on the intersection of teacher engagement and DMC. To minimise false negatives, we did not hard-code specific measurement-related terms into the search strings. Instead, we employed proximity operators to tightly couple teacher terms with specific engagement constructs, and to constrain multimodal terms to composing contexts, ensuring that the semantic focus remained strictly on writing practices rather than general technology use.

The Teacher Block required "teacher" to appear in close proximity to mechanism-related terms capturing engagement, enactment, and cognition. This list included core constructs (e.g., fidelity, self-efficacy, TPACK) and broad phrasings identified through sensitivity checks (e.g., perspectives, feedback practices) to ensure comprehensive retrieval of qualitative evidence. The DMC Block constrained "digital multimodal" or "multimodal" to appear near "composing" or "writing". Specific field terminologies (e.g., "digital storytelling", "DMC") were also included, strictly controlled by proximity operators to exclude off-topic acronyms.

We queried four major bibliographic databases: Web of Science

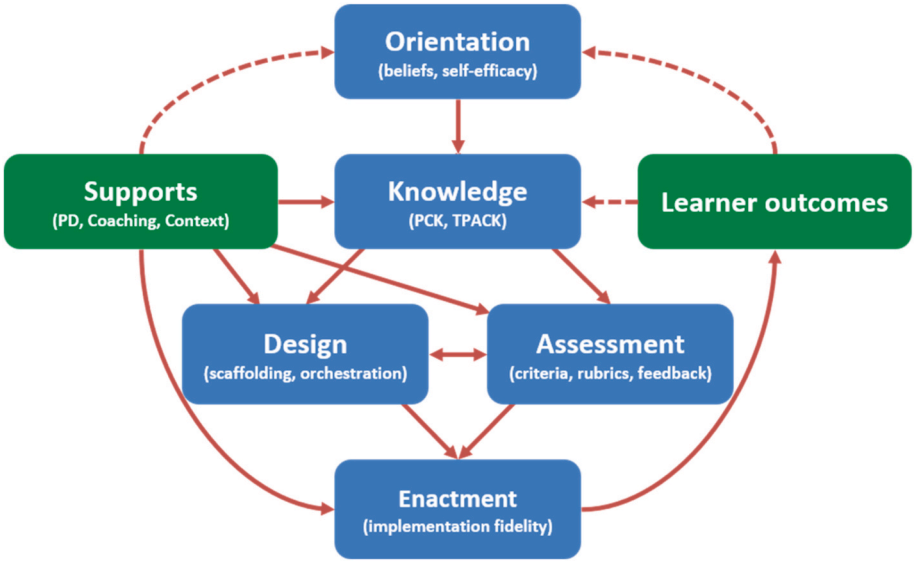


Fig. 1. The OK-DAE Framework of Teacher Engagement.
Note. Solid arrows denote direct enablers or foundational antecedents, whereas dashed arrows indicate indirect influences and reflective feedback loops. Core OK-DAE dimensions are presented in blue, while contextual and outcome layers are presented in green.

(WoS) Core Collection, Scopus, EBSCOhost (Academic Search Premier/Education Research Complete), and ProQuest (ERIC). Searches were conducted in September 2025, limited to the period 2015–2025, English-language publications, and peer-reviewed articles or reviews. Grey literature (e.g., theses, preprints) was excluded. We preserved Boolean structure, truncation, phrase searching, and proximity, adapting field scopes and operators per database. To ensure reproducibility, the exact search strings and platform-specific limits are detailed in Table 1.

3.3. Eligibility criteria

To align our screening and coding procedures with the research questions, we operationalised the study's key constructs through specific inclusion and exclusion criteria. For teacher engagement, we defined this as explicit, auditable evidence of teachers' involvement in DMC implementation, indicated by (a) classroom behaviours (e.g., orchestration, scaffolding, feedback, assessment practices), (b) design or assessment artefacts coded with defined criteria, or (c) belief-oriented constructs (e.g., attitudes, goals, teacher self-efficacy) assessed with documented instruments. Furthermore, we excluded (i) studies that only report student perceptions; (ii) purely narrative teacher reflections without systematic coding; (iii) measures unrelated to DMC enactment; (iv) preservice teachers' intentions or attitudes without actual classroom implementation; and (v) technology-acceptance-type studies (e.g., TAM, UTAUT) that are not situated in classroom-embedded DMC tasks.

Learner outcomes were defined as observable student-level results within DMC tasks, including (a) language-production quality (e.g., complexity, accuracy, lexical diversity, fluency indices [CALF]), (b) multimodal product quality (rubric-based), (c) engagement or motivation (validated scales or logged behaviours), (d) writing achievement, grades or performance scores, and (e) speaking or oral performance in multimodal tasks. We excluded preference or satisfaction reports without validated measures. When course or teacher-assigned grades were used, inclusion required evidence of independent scoring and/or inter-rater reliability (or a documented moderation process); otherwise, such outcomes were treated as background and excluded from the linkage synthesis.

To capture how teacher engagement connects to learner outcomes, we restricted our scope to instructional tasks that orchestrated linguistic and non-linguistic modes (e.g., digital storytelling [DST], video composition, podcasts, embodied multimodal composing [EMC]), embedded in instructional contexts rather than standalone tool demonstrations. We identified three linkage types to capture the analytical connections between the two sides. *DirectLink* refers to studies that reported statistically tested learner outcomes within DMC instruction and provided explicit teacher-engagement evidence (e.g., documented instructional practices) relevant to the DMC implementation. *MappedLink* denotes studies where teacher–learner relations were qualitatively inferred or narratively described without quantitative tests of teacher–learner linkage. *NoLink* applies to studies that reported teacher or learner measures independently, without an analytical bridge between the two.

3.4. Study selection process

The study selection process followed the PRISMA 2020 guidelines (Page et al., 2021; Fig. 2). The initial search identified a total of 510 records (WoS = 122; Scopus = 150; EBSCO = 139; ProQuest = 99). After merging records in Zotero and deduplicating by DOI (with a composite key fallback of title, author, and year), 288 duplicates (56.5%) were removed, leaving 222 unique records for screening.

Title and abstract screening was conducted independently by two researchers using a three-category decision system (Include/Maybe/Exclude). This process yielded a Cohen's $\kappa = .82$, indicating near-perfect agreement. Seventeen disagreements were adjudicated by an independent third reviewer, resulting in 63 studies proceeding to full-text review.

For the full-text assessment, to ensure reliability, the same two researchers independently screened a random subsample of 13 studies (20.6% of the full texts). Inter-rater agreement was substantial (Cohen's $\kappa = .63$), calculated based on the initial independent decisions, with only one disagreement identified across the full-text subsample, which was resolved via documented adjudication. Of the 63 full-text articles assessed, 12 were excluded for the following reasons: lack of systematic teacher measurement or participation ($n = 6$); theoretical or review nature without empirical data ($n = 2$); absence of measurable learner outcomes ($n = 2$); full-text unavailability ($n = 2$). Consequently, 51 empirical studies met all inclusion criteria and were retained for synthesis.

Concurrently with the full-text screening, we conducted a Risk of Bias (RoB) assessment to address the inherent subjectivity in grading

Table 1
Search strings and database parameters.

Database	Limits	Search String
Web of Science Core Collection	Topic; 2015–2025; English; Article/ Review	TS=((teacher* NEAR/3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR percept* OR perspect*)) AND ((“digital multimodal” NEAR/3 (compos* OR writ*)) OR (multimodal NEAR/3 (compos* OR writ*)) OR “digital storytelling” OR (DMC NEAR/3 (compos* OR writ* OR multimodal))))
Scopus	TITLE-ABS-KEY; 2015–2025; English; Article/ Review	TITLE-ABS-KEY(teacher* W/3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR perspect* OR percept*) AND ((“digital multimodal” W/3 (compos* OR writ*)) OR (multimodal W/3 (compos* OR writ*)) OR “digital storytelling” OR (DMC W/3 (compos* OR writ* OR multimodal))))
EBSCOhost (Academic Search Premier/ Education Research Complete)	TI/AB/SU; Peer Reviewed; 2015–2025; English; Article/ Review	((TI teacher* AND TI(teacher* N3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR perspect* OR percept*)) AND ((TI “digital multimodal” N3 (compos* OR writ*)) OR (TI multimodal N3 (compos* OR writ*)) OR TI “digital storytelling” OR (TI DMC AND TI (compos* OR writ* OR multimodal)))) OR (AB teacher* AND AB(teacher* N3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR perspect* OR percept*)) AND ((AB “digital multimodal” N3 (compos* OR writ*)) OR (AB multimodal N3 (compos* OR writ*)) OR AB “digital storytelling” OR (AB DMC AND AB (compos* OR writ* OR multimodal)))) OR (SU teacher* AND SU(teacher* N3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR perspect* OR percept*)) AND ((SU “digital multimodal” N3 (compos* OR writ*)) OR (SU multimodal N3 (compos* OR writ*)) OR SU “digital storytelling” OR (SU DMC AND SU (compos* OR writ* OR multimodal))))
ProQuest (ERIC)	NOFT; Peer Reviewed; 2015–2025; English; Article/ Review	(teacher* NEAR/3 (engag* OR participat* OR invest* OR enact* OR implement* OR fidel* OR belief* OR attitude* OR self-efficac* OR PCK OR TPACK OR orchestr* OR scaffold* OR assess* OR feedback* OR perspect* OR percept*)) AND ((“digital multimodal” NEAR/3 (compos* OR writ*)) OR (multimodal NEAR/3 (compos* OR writ*)) OR “digital storytelling” OR (DMC NEAR/3 (compos* OR writ* OR multimodal)))

Note. TS = Topic; TI = Title; AB = Abstract; SU = Subject terms; NOFT = No full text field; NEAR/n, W/n, Nn = Proximity operators (within n words).

DMC. Specifically, we systematically evaluated rater subjectivity (*RoB_Rater*) to determine whether the included studies mitigated grading bias through independent double-scoring, blinded reviews, or reported inter-rater reliability metrics. Studies relying on single-rater assessment or lacking explicit reliability indicators were formally flagged as high risk. This integrated appraisal allowed us to critically evaluate the reliability of the reported learner outcomes while retaining valuable exploratory data.

3.5. Coding framework and reliability

Data extraction and coding were conducted using a standardised protocol where the study served as the unit of analysis. When a paper reported multiple cohorts or tasks, we coded each contrast separately but aggregated them to the study level for narrative synthesis, unless comparability permitted a meta-analysis. Regarding measurement priority, we prioritised validated instruments for direct measurement; where multiple indicators existed, we retained all data points but distinguished between validated scales and ad hoc measures for analysis. In terms of linkage classification, where studies reported both qualitative and quantitative links, both were coded, with the primary analytic pathway labelled according to the study's dominant methodological approach. Finally, for overlapping samples across papers, we treated these as one study unit by prioritising the most complete report.

The coding of contextual factors (e.g., task complexity, AI involvement) followed established precedents in DMC research (e.g., Hafner, 2015; Lim & Polio, 2020) and systematic review methodology (e.g., Meader et al., 2016). The comprehensive coding framework for teacher engagement and learner outcomes, along with representative examples, is presented in Table 2. The contextual, design, and linkage variables are detailed in Table 3.

Prior to formal data extraction, two independent researchers underwent a training and calibration session to align their understanding of the codebook. Both coders first independently evaluated a pilot sample of five studies, resolving initial discrepancies through discussion to refine the rubrics. Subsequently, they independently coded a random subsample of 10 studies (approximately 20% of the included corpus) to establish inter-coder reliability. This yielded a Cohen's κ of .79, indicating substantial agreement. The remaining studies were then independently double-coded. Disagreements were resolved via consensus-building discussions, with a third experienced researcher acting as an independent adjudicator for any unresolved items.

3.6. Data Analysis

All analyses were conducted on the cleaned dataset, and all visualisations were produced in R (v. 4.5.1). Descriptive patterns were examined through four figures, constructed using co-occurrence and descriptive-mapping procedures adapted from Chen and Song (2019). These procedures enabled a systematic characterisation of how teacher engagement has been operationalised across the included studies.

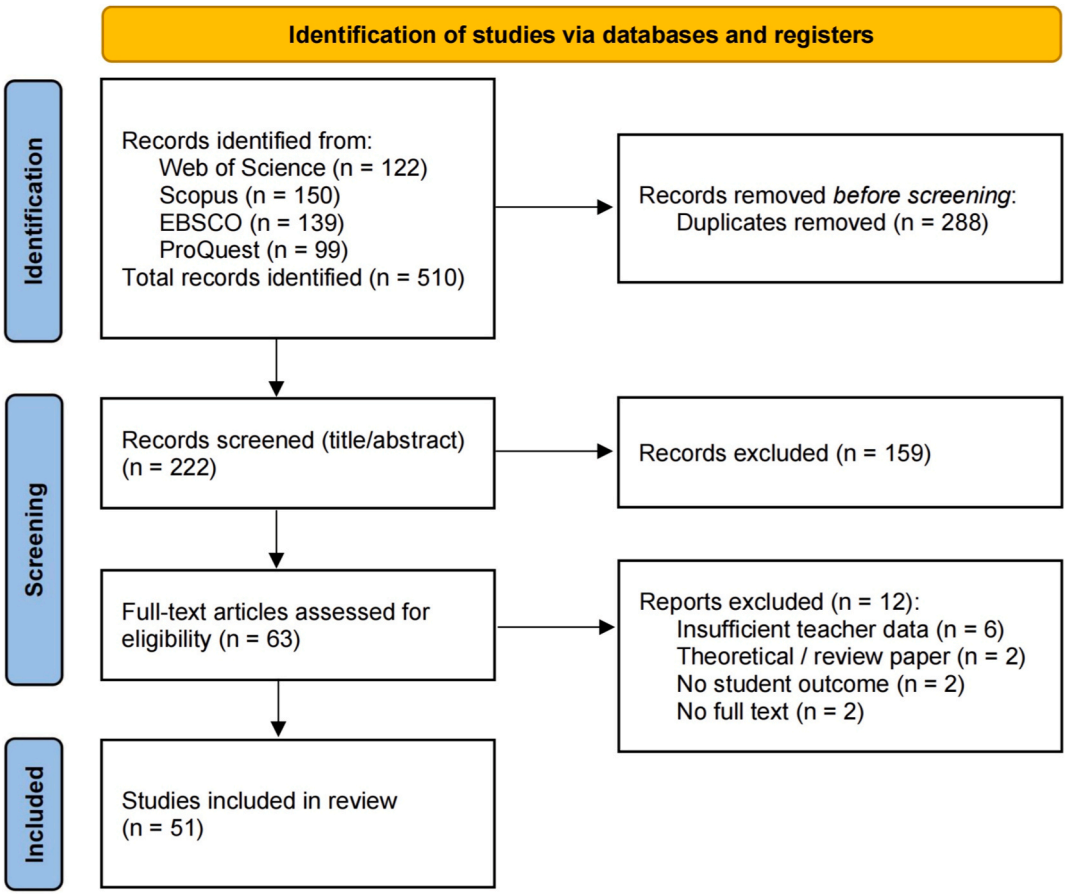


Fig. 2. PRISMA 2020 flow diagram of study selection.

4. Results

The results are presented in three main sections to foreground the central empirical challenge identified in this review. Given that the ultimate pedagogical goal of DMC is to facilitate student learning, we begin our synthesis by mapping the empirical evidence concerning the linkages between teacher engagement and learner outcomes (RQ3). Second, we provide a descriptive overview of the included studies (RQ1) to contextualise the field's geographic, educational, and methodological features. Finally, we map the co-occurrence patterns among teacher engagement dimensions (RQ2) to reveal how instructional practices cluster within the literature.

4.1. Teacher-learner linkages

4.1.1. Study design and linkage type

As shown in Fig. 3, qualitative designs dominated the corpus ($n = 42$, 82.4%), followed by mixed-methods ($n = 8$, 15.7%) and a single experimental study ($n = 1$, 2.0%). Within this dominant qualitative paradigm, exploratory qualitative designs ($n = 19$) and case studies ($n = 16$) were the most frequently used approaches, while iterative approaches to pedagogical development, including Action Research ($n = 5$) and Design-Based Research ($n = 2$), were the least represented. This reflects the interpretive tradition of DMC research, emphasising contextualised inquiry over controlled experimentation.

Building on these design characteristics, we then report the distribution of linkage types between teacher and learner perspectives. For data linkage between teacher and student perspectives, MappedLink ($n = 32$) was the most frequent type, indicating inferred associations; NoLink ($n = 17$) separated teacher and student data; and only two studies provided DirectLink evidence via statistically tested learner outcomes alongside explicit teacher-engagement evidence. These findings reveal that although multi-perspective data collection is common, direct integration of teacher-engagement evidence with statistically tested learner outcomes remains rare.

Furthermore, regarding methodological rigour, the quality appraisal indicated that 41 out of the 51 included studies (80.4%) exhibited a high risk of rater bias, relying on single-rater assessments or lacking explicit reliability indicators. In contrast, only 10 studies (19.6%) reported rigorous validation procedures such as double-blind scoring or explicit reliability indices (e.g., Cohen's Kappa or percentage agreement; see Simsek, 2020; Wang et al., 2024).

4.1.2. Study-level dimensional mapping

Fig. 4 visualises the relationship between the number of teacher- and learner-side dimensions measured across studies. Each point represents one study, with colour indicating the linkage type (NoLink, MappedLink, or DirectLink). The horizontal and vertical axes correspond to the number of learner- and teacher-side dimensions, respectively. Most studies clustered around 2–3 teacher dimensions but only 0–1 learner dimension(s), with few achieving both high-dimensional coverage and DirectLink evidence. Specifically, the teacher-side mean = 2.39 (range 1–4), while the learner-side mean = .88 (range 0–3). Among the 51 studies, 32 were MappedLink, 17 were NoLink, and only 2 were DirectLink, indicating that outcome-tested connections alongside explicit teacher-engagement evidence remain rare.

This pattern shows that most studies adopted descriptive rather than integrative approaches connecting teacher engagement with learner outcomes in the current DMC literature. Only two studies in the corpus employed experimental or quasi-experimental logic to connect instructional enactment with learner outcomes: one adopted a pretest–posttest quasi-experimental design (Arroba & Acosta, 2021) and the other,

Table 2
The coding framework and representative examples for teacher engagement and learner outcomes.

Code	Operational Definition	Representative Examples
Teacher Engagement		
ORI	Orientation: Beliefs, motivations, and self-efficacy.	• Measuring pedagogical beliefs and attitudes toward authentic DST through targeted surveys (Arroba & Acosta, 2021). • Capturing design orientations via collaborative co-design sessions through a transliteracies lens (James & McCarthey, 2024). • Categorising teachers' engagement into incidental, ambivalent, and integral approaches (Jiang et al., 2021).
KNO	Knowledge: Knowledge bases (e.g., PCK, TPACK).	• Leveraging TPACK to link Snapchat affordances with specific literary analysis practices (Bender, 2020).
DES	Design: Scaffolding, orchestration practices, and tool selection.	• Designing instructional frameworks using ArcGIS StoryMaps to integrate critical literacy and local history (Purnomo & Chen, 2025). • Applying SFL metafunctions to structure and evaluate digital literacy tasks (Wang et al., 2024). • Designing learning experiences with a pedagogic metalanguage to support semiotic meaning-making (Lim & Unsworth, 2023).
ASM	Assessment: Use of criteria, rubrics, and feedback routines.	• Developing translanguaging assessment rubrics to link abstract curricular constructs to students' lived experiences (Jiang et al., 2024). • Employing rubric-based assessments evaluating organisation, language quality, and central message delivery (Arroba & Acosta, 2021). • Using joint tech-enhanced retelling with rubric-based peer and teacher assessment (Simsek, 2020).
ENA	Enactment: Implementation fidelity and classroom delivery.	• Scaffolding an eight-lesson process-approach intervention for digital story-writing using Google Slides (Cheung, 2024). • Implementing a year-long program focusing on the strategic use of multiple semiotic resources (Jiang et al., 2021). • Facilitating student multimodal composing within a SPOC environment to guide meaning-making (Wang et al., 2024).
Learner Outcomes		
CALF	Complexity, Accuracy, Lexical Diversity, and Fluency: Linguistic indices of writing quality.	• Comparing CAF through grammatical accuracy analysis of digital narratives (Simsek, 2020).
MPQ	Multimodal Product Quality: Quality assessed via rubrics or expert ratings.	• Evaluating video documentaries and eBooks as evidence of conceptual representation and longitudinal learning (Jiang et al., 2024). • Assessing students' semiotic awareness and multimodal literacy through created artefacts (Lim & Unsworth, 2023).
ENG	Engagement: Motivation, participation levels, and logged behaviours.	• Profiling student engagement into ideational, interpersonal, and textual interactions within multimodal forums (Wang et al., 2024). • Capturing student engagement through qualitative mapping of interviews, artefacts, and classroom interactions (James & McCarthey, 2024). • Identifying cognitive, emotional, and behavioural engagement through stored logs and forum interactions (Mohammadi Zenouzagh et al., 2023).
ACH	Achievement: Course grades, test scores, or overall performance.	• Evaluating the impact of a process-writing approach on reducing grammatical mistakes in student writings (Cheung, 2024). • Measuring writing quality based on language, content, and organisation in CMC environments (Mohammadi Zenouzagh et al., 2023).
SPE	Speaking: Oral narrative ability in multimodal tasks.	• Assessing improvements in oral skills through pre-/post-tests following an authentic DST intervention (Arroba & Acosta, 2021). • Assessing significant gains in speaking performance, specifically targeting storyline and pronunciation (Purnomo & Chen, 2025).

Note. TPACK = Technological Pedagogical Content Knowledge; PCK = Pedagogical Content Knowledge; DST = Digital Storytelling; SFL = Systemic Functional Linguistics; SPOC = Small Private Online Course; CAF = Complexity, Accuracy, and Fluency; CMC = Computer-Mediated Communication.

Table 3
The coding framework for contextual, design, and linkage variables.

Category	Code	Operational Definition
CONTEXT & DESIGN	LEVEL	Educational Level: Primary, Secondary, Tertiary, or Cross-level.
	CONTEXT	Language Context: L1, L2 (ESL/EFL), or Mixed language environments.
	Country_Region	Country/Region: Geographic location where the study was conducted.
	Task_Complexity	Task Complexity: Single task type (1) vs. Multiple task types (2).
	AI_Involvement	AI Involvement: Non-AI tasks (0) vs. tasks involving GenAI tools (1).
	Study_Design	Study Design: Qualitative, Mixed Methods, or Experimental/Quasi-experimental.
LINKAGE	Link_Type	Linkage Type: DirectLink, MappedLink, or NoLink.

Note. L1 = First Language; L2 = Second Language; ESL = English as a Second Language; EFL = English as a Foreign Language; GenAI = Generative Artificial Intelligence.

formally classified as mixed methods, embedded an experimental comparison within its procedure (Cheung, 2024). In both cases, learner outcomes were statistically tested within a DMC intervention, and teacher engagement was evidenced through documented instructional enactment and implementation features rather than a standalone measured construct.

4.2. Descriptive overview of the dataset

4.2.1. Geographical distribution

Fig. 5 maps the global distribution of studies and provides the exact quantitative breakdown of included studies per country. A total of 54 countries were mentioned in 51 studies, because several cross-context studies collected data from multiple countries or regions. Most

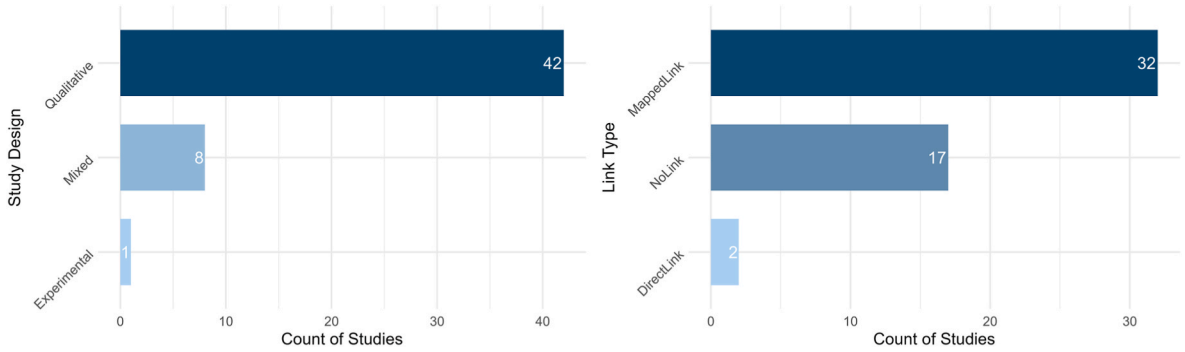


Fig. 3. Study design and linkage type distribution ($N = 51$).

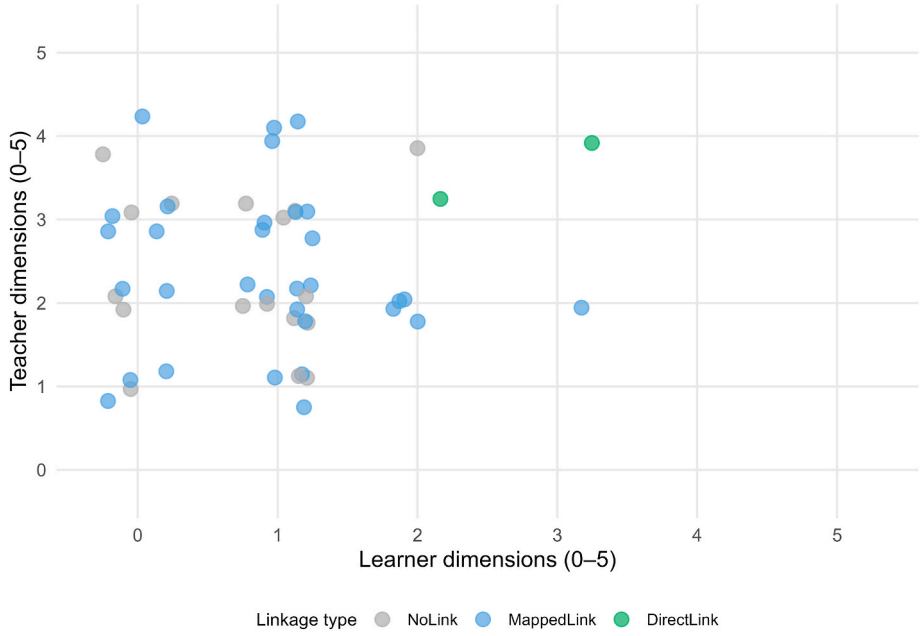


Fig. 4. Study-level relationship between teacher and learner dimensions ($N = 51$).

originated from the United States ($n = 13$), Mainland China ($n = 11$) and Hong Kong SAR ($n = 7$), followed by Australia ($n = 5$), and Canada ($n = 4$). Consequently, the dataset is geographically uneven, predominantly concentrated in Anglophone and Sinophone contexts, while research from other regions remains limited.

4.2.2. *Educational level and learning context*

As shown in Fig. 6, most studies on teacher engagement in DMC were conducted at the secondary level ($n = 23$, 45.1%), followed by tertiary ($n = 18$, 35.3%) and primary ($n = 8$, 15.7%), with only two cross-level designs (one primary and secondary; one secondary and tertiary). This pattern indicates that DMC pedagogy has primarily been implemented with adolescents and young adults, consistent with its frequent adoption in secondary and tertiary educational settings. Regarding learning context, the majority of studies were situated in EFL environments ($n = 14$), followed by L1-English ($n = 12$), multilingual/mixed ($n = 9$), ESL ($n = 8$), and non-English settings ($n = 8$). Together, EFL and L1-English contexts accounted for the largest share of studies in the dataset.

4.2.3. *Task complexity and AI involvement*

Regarding task design, the reviewed studies encompassed a wide spectrum of DMC formats. While the specific tasks ranged from DST and video composition to more specialised genres such as podcasts, eBooks, and EMC, these diverse instructional activities were analytically unified under the DMC construct. These tasks leveraged various digital platforms with distinct affordances, including cloud-based collaborative tools (e.g., Google Slides), social media (e.g., Snapchat), and geospatial software (e.g., ArcGIS StoryMaps). Selected representative examples of these typologies, alongside their associated instructional and assessment strategies, are detailed in Table 2.

Fig. 7 displays two complementary dimensions of task design. Most studies adopted single-task designs ($n = 39$, 76.5%), while multi-task implementations accounted for 23.5% ($n = 12$). This imbalance shows that, although DMC tasks are often conceptualised as process-oriented, most of the systematically reviewed studies still evaluated one-off products rather than extended composing sequences. With respect to technological mediation, AI-enhanced DMC research remains in its infancy: only two studies (3.9%) explicitly integrated GenAI tools during composing. The remaining 49 studies (96.1%) used conventional digital tools such as video-editing or presentation software. This indicates that AI-enhanced DMC remains rare within the current literature. Together, these descriptive findings establish a comprehensive quantitative portrait of the existing DMC-teacher engagement literature, providing the empirical basis for the co-occurrence analyses.

4.3. *Mapping co-occurrence of teacher and learner dimensions*

To move beyond descriptive mapping, this subsection examines co-

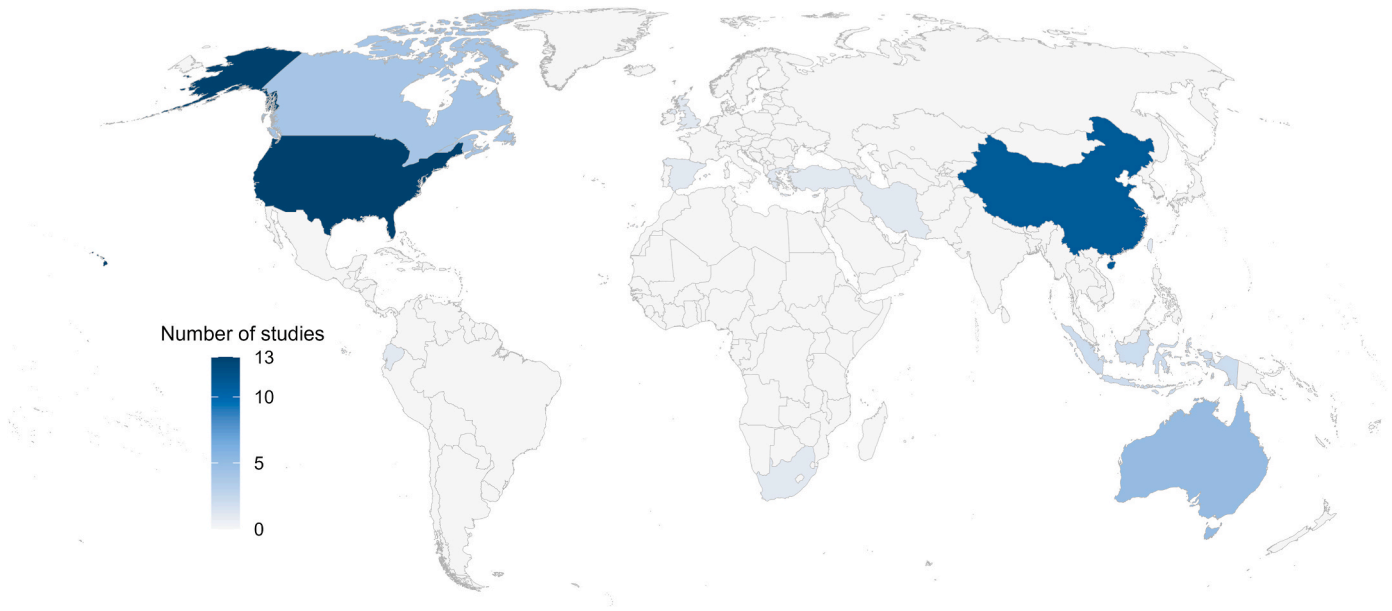


Fig. 5. Geographical Distribution of Studies ($N = 51$; 54 country/region mentions).
Note. The exact number of studies per country/region is as follows: United States ($n = 13$), Mainland China ($n = 11$), Hong Kong SAR ($n = 7$), Australia ($n = 5$), Canada ($n = 4$), Indonesia ($n = 2$), Singapore ($n = 2$), Cyprus ($n = 1$), Ecuador ($n = 1$), Greece ($n = 1$), Iran ($n = 1$), Norway ($n = 1$), South Africa ($n = 1$), Spain ($n = 1$), Taiwan ($n = 1$), Turkey ($n = 1$), and United Kingdom ($n = 1$).

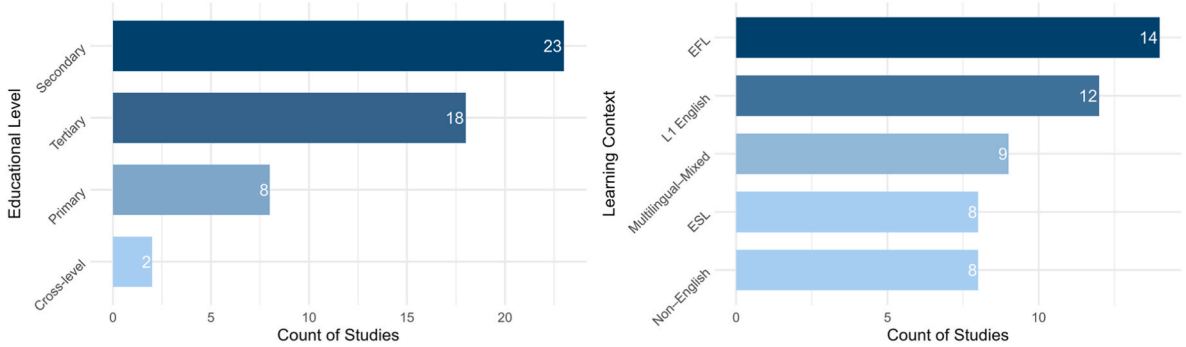


Fig. 6. Educational level and learning context distribution ($N = 51$).

occurrence patterns among teacher- and learner-related variables. Following the OK-DAE framework, each study was assessed for the presence of five teacher-engagement dimensions (Orientation, Knowledge, Design, Assessment, and Enactment) and five learner-outcome dimensions (CALF, MPQ, Engagement, Achievement, and Speaking). When studies exhibited more than two components, all identified dimensions were independently coded and contributed to multiple pairwise connections within the matrix. Heatmaps were generated to visualise co-occurrence patterns for the full dataset (teacher-side co-occurrence) and for the subset coded as MappedLink/DirectLink (teacher–learner co-occurrence) to provide holistic and coded-connection-specific perspectives. This approach, common in systematic reviews (Chen & Song, 2019; Meader et al., 2016), identifies specific variable combinations that frequently cluster together across the empirical literature.

4.3.1. Teacher-side co-occurrence patterns

Across the 51 included studies, Design was the most frequently observed dimension ($n = 41$), followed by Enactment ($n = 38$), Assessment ($n = 22$), and Orientation ($n = 20$), whereas Knowledge was rarely explicitly identified ($n = 1$). As shown in Fig. 8, teacher-engagement components exhibited markedly uneven co-occurrence patterns across the 51 studies. The strongest overlap occurred between Design and Enactment (33 studies), indicating that studies on DMC teacher engagement often treat DMC practice as a design-to-implementation continuum. Other frequent pairings included Enactment–Assessment (17), Design–Assessment (16), Design–Orientation (16), and Orientation–Enactment (15). In contrast, Knowledge co-occurred with few other components (0–1 times each). Overall, the pattern highlights a concentration of research around pragmatic engagement, how teachers design, assess, and carry out multimodal tasks, rather than conceptual or knowledge-based engagement that integrates intention, expertise, and evaluation.

4.3.2. Teacher–learner cross-co-occurrence patterns

Fig. 9 presents the cross-matrix of teacher- and learner-related variables within the subset of studies that reported some form of teacher–learner connection (MappedLink/DirectLink, $N = 34$). The strongest intersections were found between Design and Engagement (18 studies) and between Enactment and Engagement (18). These patterns indicate that these instructional dimensions were the most examined in relation to learners' participation and multimodal engagement. Additionally, Orientation–Engagement (8) and Assessment–Engagement (8) links were also identified. Moderate overlaps occurred between Design/

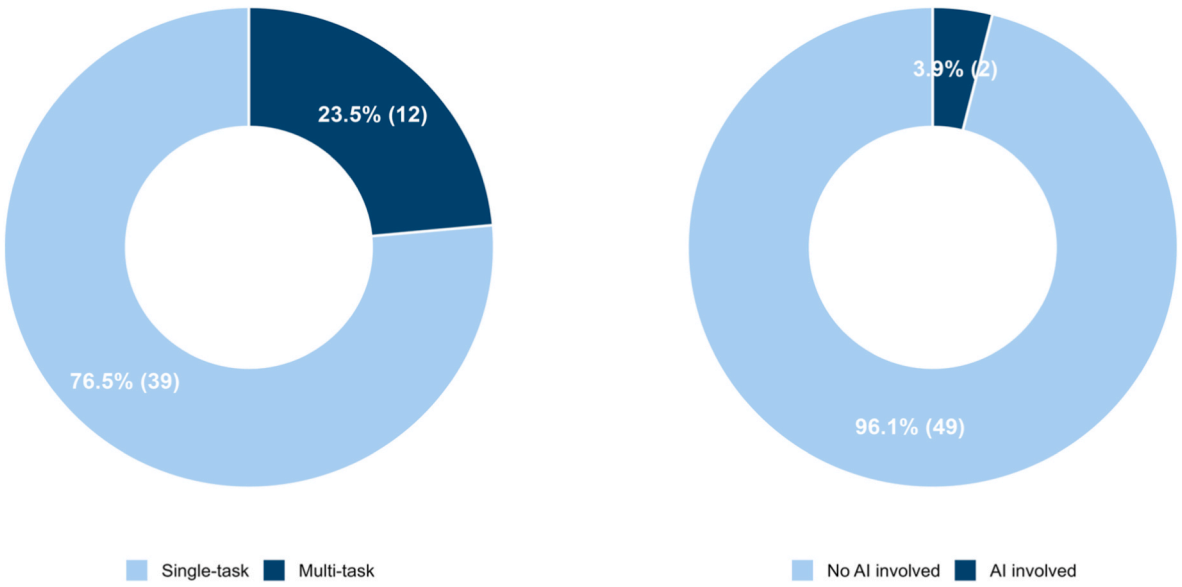


Fig. 7. Task complexity and AI involvement distribution ($N = 51$).

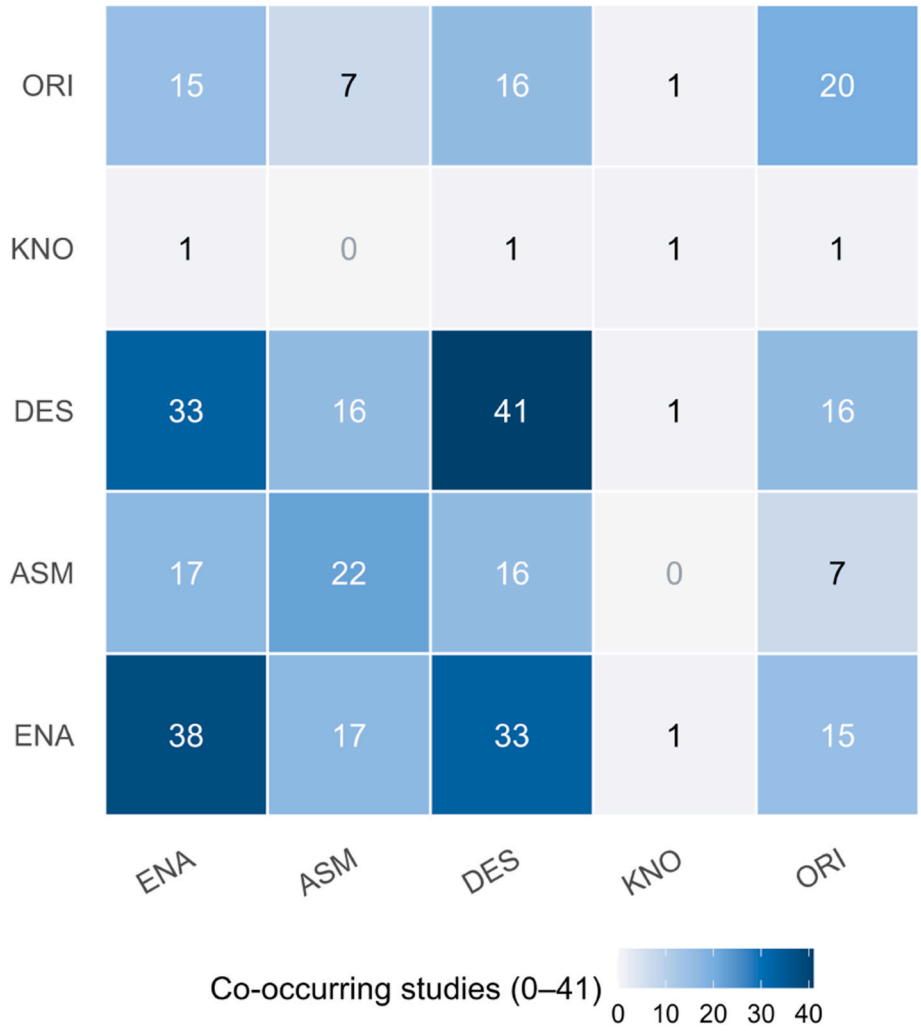


Fig. 8. Co-occurrence among teacher engagement components (all studies, $N = 51$).

Enactment and MPQ (4 each), and between Assessment and Achievement (4).

Connections with performance-based outcomes such as Speaking and CALF were marginal (≤ 3 and ≤ 1, respectively), and Knowledge was

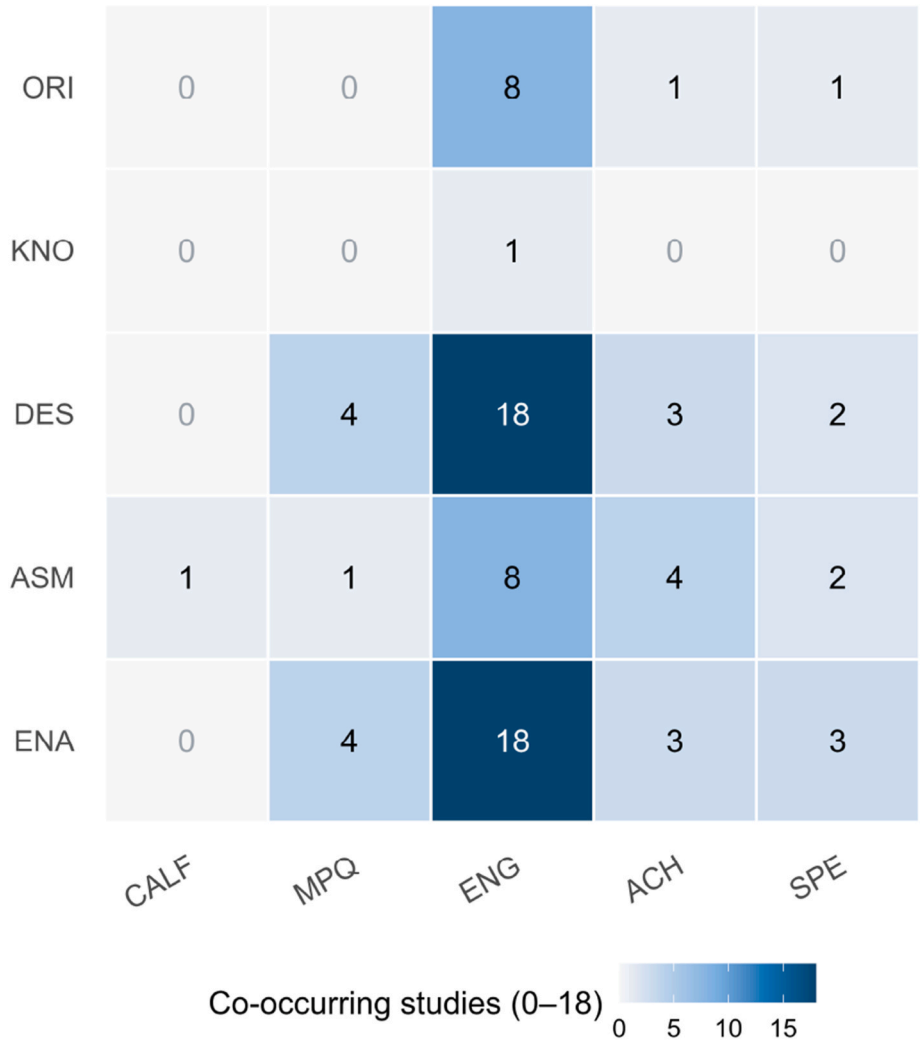


Fig. 9. Teacher–Learner co-occurrence (Linked evidence only, $N = 34$).

again rarely linked to any learner-side dimension. Taken together, these teacher–learner intersections show that current evidence mainly relates to co-occurrence between instructional variables and learner engagement measures. Overall, existing research most frequently examines teacher engagement alongside students' engagement, with more limited links to performance-based outcomes (e.g., CALF, Speaking).

5. Discussion

This review mapped 51 empirical studies (2015–2025) examining teacher engagement in DMC across diverse educational and linguistic contexts. Methodologically, the corpus relied heavily on qualitative designs and rarely established explicit analytic linkages between teacher engagement variables and learner outcomes. The descriptive analyses indicate that the existing evidence base is concentrated primarily in secondary and tertiary L2 settings and geographically clustered in Anglophone and Sinophone regions. A large proportion of studies focused on single, one-off multimodal products rather than extended composing sequences, suggesting an emphasis on outcomes rather than processes. Taken together, these patterns delineate the empirical contours of the field and provide the basis for the interpretations that follow.

5.1. Methodological Asymmetry and the "Last-mile" problem

As foregrounded in the empirical findings, the most critical structural issue identified in this review is a pronounced methodological asymmetry regarding the connection between teacher engagement and learner outcomes. We conceptualise this scarcity of empirical linkage not merely as an absence of evidence, but as a systemic "last-mile" problem in the field. While the literature is rich in qualitative descriptions of classroom practice and teacher perceptions, few studies integrate teacher-engagement evidence with statistically tested learner outcomes in ways that permit analytic linkage.

Specifically, though methodologically fragmented, the empirical findings indicate recurring patterns in how teacher engagement is associated with learner outcomes, particularly in studies that document teacher practices without modelling their relationship to outcomes. For example, incorporating human-centred design challenges that resonate with students' out-of-school multimodal practices has been shown to positively shift their attitudes toward writing (James & McCarthey, 2024). Similarly, when teachers use DMC as a formative assessment tool, it enables connections between abstract curricular concepts and students' lived experiences (Jiang et al., 2024). Targeted teacher feedback, such as guidance on image selection and storyline development, has been associated with improved emotional regulation during the composing process (Purnomo & Chen, 2025).

By contrast, a small number of studies explicitly link instructional

practices to measurable outcomes. For instance, implementing a process approach to digital story-writing, where teachers actively scaffold planning and revision through collaborative tools, has been associated with higher student engagement and fewer grammatical errors (Cheung, 2024). Likewise, the use of structured, rubric-based assessments for authentic DST has been linked to statistically tested gains in learners' oral proficiency (Arroba & Acosta, 2021). Similarly, structured peer and teacher assessment routines have been shown to directly impact learners' linguistic complexity, accuracy, and fluency (CAF) in collaborative digital narratives (Simsek, 2020). Furthermore, variations in instructional modalities have been analytically linked to shifts in cognitive and emotional engagement as well as overall writing achievement (Mohammadi Zenouzagh et al., 2023).

However, these findings remain illustrative rather than conclusive, as teacher engagement is rarely operationalised as an independent construct to explain variance in learner outcomes. Even in the explicit linkage studies, the evidential chain remains limited: outcomes are quantified at the level of the overall intervention, with limited analytical separation of engagement components. As a result, it remains difficult to determine which specific dimensions of teacher engagement are most consequential for distinct learner outcomes. Our quality appraisal further confirms a systemic methodological vulnerability regarding their reliability. The majority of the reviewed studies (over 80%) rely primarily on single-teacher assessments or unstandardised rubrics rather than rigorous inter-rater validation, showing a high risk of rater bias. This widespread subjectivity in assessing learner outcomes constitutes a significant barrier to establishing robust teacher-learner linkages.

This gap highlights a persistent "last-mile" problem in DMC research: the field increasingly documents what teachers do, yet rarely provides the methodological infrastructure needed to test how specific dimensions of teacher engagement translate into verifiable learner gains. More broadly, many studies rely on self-reported perceptions, reflective accounts, or single-instance artefact analyses, often without parallel learner-outcome measures that would permit linkage testing, and may blur the distinction between perceived instructional innovation and demonstrable learning effects.

Several factors likely sustain this methodological imbalance. Institutional and resource constraints often limit opportunities for longitudinal data collection, particularly in under-resourced EFL contexts. In addition, the rapid expansion of digital tools, including GenAI, complicates attribution, as instructional work and cognitive labour are increasingly distributed across teachers, learners, and technologies. Consequently, the current evidence base remains largely descriptive and intervention-centred rather than explanatory, constraining cumulative theory-building and leaving the pathways between teacher engagement and learner outcomes more often assumed than empirically demonstrated.

5.2. Structural imbalance in teacher engagement with DMC

This review reveals a pronounced structural imbalance in how teacher engagement with DMC has been operationalised in the empirical literature. Across the reviewed studies, engagement is predominantly evidenced through observable instructional practices, most notably task design, classroom scaffolding, and enactment, while less visible dimensions such as teachers' knowledge bases and assessment practices receive comparatively limited attention.

Studies such as Elola et al. (2025) and Jiang et al. (2021) exemplify this tendency. Elola et al. (2025) provide a granular account of collaborative task design and implementation in a Spanish L2 course, foregrounding teachers' and students' perceptions. Similarly, Jiang et al. (2021) examine teacher engagement in a Chinese tertiary EFL context, revealing that teachers often operationalise DMC as a technical instrument for traditional language drills while giving only incidental attention to multimodal assessment. Both studies illustrate how research prioritises the description of enactment processes over the explication of the evaluative reasoning used to judge learner success.

This unidimensional focus reflects a broader privileging of observable performance over latent cognition. While visible Enactment is easier to document and analyse, marginalising dimensions such as Knowledge and Assessment risks obscuring the complex expertise required to sustain high-quality DMC instruction. For example, rarely do studies explicitly capture how a teacher's TPACK shapes their ability to leverage digital affordances for domain-specific tasks, such as literary analysis (Bender, 2020). Moreover, designing tasks that transcend mere technical production requires a deep grounding in pedagogic metalanguage (Lim & Unsworth, 2023) or Systemic Functional Linguistics (SFL; Wang et al., 2024) to guide and assess students' semiotic meaning-making effectively. Evidence from professional learning research similarly suggests that behavioural enactment alone rarely leads to durable pedagogical change unless supported by deeper cognitive, motivational, and knowledge-based foundations (McNaughton et al., 2018; Sims & Fletcher-Wood, 2021). In this sense, strong Enactment without corresponding Knowledge development or Assessment literacy is more likely to result in short-term activity than in sustained instructional improvement.

Taken together, these patterns suggest that teacher engagement in DMC has been conceptualised narrowly, limiting the field's capacity to examine engagement as a set of interdependent dimensions spanning Orientation, Knowledge, Design, Assessment, and Enactment. Currently, because empirical studies rarely adopt a holistic approach that investigates more than two or three dimensions simultaneously, a robust comparison of instructional outcomes between comprehensive and narrowly focused interventions remains unfeasible. A more integrative perspective is therefore needed to capture how these dimensions interact to support sustained teacher engagement over time.

5.3. Limitations

Several limitations must be acknowledged when interpreting the findings of this review. First, regarding the search and selection process, despite employing rigorous, pre-registered screening protocols on OSF to ensure robustness, the final corpus is relatively small ($n = 51$) and predominantly comprises studies published in English. Furthermore, within this corpus, the subset of studies establishing quantitative teacher-learner linkages (DirectLink) is particularly scarce. This geographic, linguistic, and empirical concentration may limit the generalisability of the findings to other educational contexts.

Second, a notable limitation lies in the OK-DAE framework itself. As currently conceptualised, the framework functions strictly as a descriptive analytic heuristic derived from existing literature. It has not yet been empirically validated as a predictive or causal model, meaning its dimensions are currently evaluative rather than explanatory.

Third, there is a systemic limitation regarding the methodological quality of the reported learner outcomes. As revealed by our RoB assessment, a vast majority of the included studies relied on single-rater subjective assessments or unstandardised rubrics without reporting inter-rater reliability. This high risk of bias complicates the ability to make robust, cross-study comparisons of pedagogical effectiveness.

Fourth, our methodological analysis reveals that the field is heavily reliant on exploratory qualitative designs and cross-sectional case studies. Iterative approaches to pedagogical development (e.g., action research, design-based research) remain underrepresented. Consequently, much of the current evidence captures single-instance interventions rather than sustained instructional change, further complicating the establishment of robust, long-term teacher-learner linkages.

6. Conclusion and implications

Synthesising evidence from 51 empirical studies (2015–2025), this review characterises a field that is expanding in scope yet remains

structurally and methodologically uneven. Structurally, the evidence base demonstrates a pronounced focus on the observable dimensions of Design and Enactment, often at the expense of less visible cognitive and evaluative dimensions such as Knowledge and Assessment. Methodologically, this unidimensional emphasis is accompanied by a persistent "last-mile" gap, in which qualitative accounts of teacher engagement are abundant, while robust analytic linkages to verifiable learner outcomes remain scarce. Taken together, the OK-DAE framework functions as a useful conceptual vocabulary for articulating these dispersed emphases. This heuristic makes visible latent connections that remain obscured in the literature, highlighting that the often under-examined dimensions of Orientation, Knowledge, and Assessment should be understood not as optional add-ons, but as core cognitive and evaluative foundations that help sustain the visible practices of multimodal instruction.

The findings of this review carry distinct implications for both professional practice and future research agendas. Pedagogically, the structural dominance of the Design and Enactment dimensions suggests that current professional development efforts may be over-emphasising technical implementation at the expense of deeper evaluative reasoning. There is, therefore, a critical need to move beyond procedural training toward cultivating multimodal assessment literacy—equipping teachers not merely to facilitate digital production, but to articulate explicit criteria for quality and to diagnose learner needs within multimodal texts. Such a shift is essential for transforming DMC from a mere technical exercise into a rigorous, assessment-driven learning experience.

Methodologically, the prevalence of descriptive, perception-based studies indicates a pressing need to bridge the "last-mile" gap between teacher engagement and verifiable learner outcomes. Future inquiry would benefit from pivoting from exploratory mapping toward explanatory testing, employing longitudinal designs and validated instruments to capture how engagement evolves beyond single-instance interventions. Specifically, moving beyond self-reported enthusiasm to model how specific engagement dimensions independently or synergistically contribute to variance in student performance would significantly strengthen the field's evidence base. To achieve this holistic perspective, future research should move beyond measuring isolated practices to develop and validate unified constructs, such as "DMC Pedagogical Literacy" or "Adaptive Multimodal Expertise". Adopting such comprehensive constructs enables researchers to capture how teachers integrate their orientations, knowledge, design, and assessment capabilities during classroom enactment. This shift would provide a cohesive empirical basis to inform teacher education.

Finally, the emerging integration of GenAI introduces new layers of complexity to the conceptualisation of teacher engagement. Although only marginally represented in the current corpus, the presence of these tools suggests that future research will need to distinguish between human-led pedagogical scaffolding and algorithmic assistance. Rather than assuming a fundamental shift in teacher roles, the challenge for the field is to refine our methodological tools so that the specific contributions of the teacher remain discernible in increasingly automated composing environments. Addressing these evolving dynamics will be essential for moving the field toward a more robust, empirically grounded understanding of teacher-learner linkages in the digital age.

CRediT authorship contribution statement

He Crane Han: Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stuart McNaughton:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Lawrence Jun Zhang:** Writing – review & editing, Visualization, Validation, Supervision, Investigation, Conceptualization.

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Data availability

Data are available in the public domain: https://doi.org/10.17605/OSF.IO/32KQ4.

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